Imaging, Positioning and SSD

the super simple study guide

Separation Worksheet #2 · Imaging and Positioning Worksheet #2
Class slides “Treatment: Imaging and Positioning”

Francine Castro · worksheets due 11/12

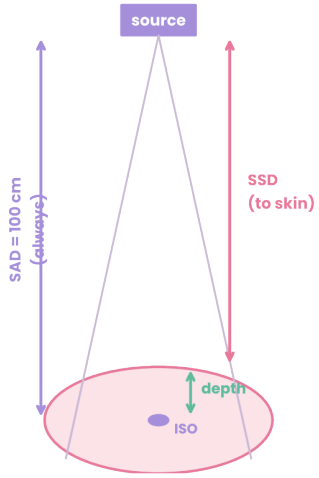
What is inside

Part	What you learn
1. SSD and SAD	The one picture that makes every SSD question easy
2. The separation trick	What to do when the ISO moves anterior, posterior, right or left
3. Separation Worksheet #2	All 7 problems, drawn and solved, with notes on your work
4. Practice	New problems with answers so you can test yourself
5. Patient positions	HFS, HFP, FFS, FFP in pictures
6. Gantry angles	TrueBeam vs Trilogy and the flip trick for RAO, LAO, RPO, LPO
7. Imaging Worksheet #2	All 9 questions, drawn, with two things to double check
8. The class slides	kV, MV, CBCT, full vs empty bladder, breath hold, belly board
9. Cheat sheet	One page to read right before the quiz

How to use this

Go slow. Do Part 2 with a pencil. Draw the bar every single time. After 3 or 4 problems your brain does it alone.

1. SSD and SAD



SAD means source to axis distance. It goes from the source to the isocenter (ISO). In your class it is **always 100 cm**. It never changes.

SSD means source to surface distance. It goes from the source to the **skin** where the beam goes in.

Depth is the space between the skin and the ISO.

So the skin sits a little before the ISO. That means:

SSD = 100 - depth

That is the only formula. Every SSD on both worksheets uses it. The hard part is only finding the right depth.

Easy picture

Think of a 100 cm tape measure that starts at the machine. The ISO is at the 100 mark. If the skin is 14 cm before the ISO, the skin is at the 86 mark. So SSD = 86 cm.

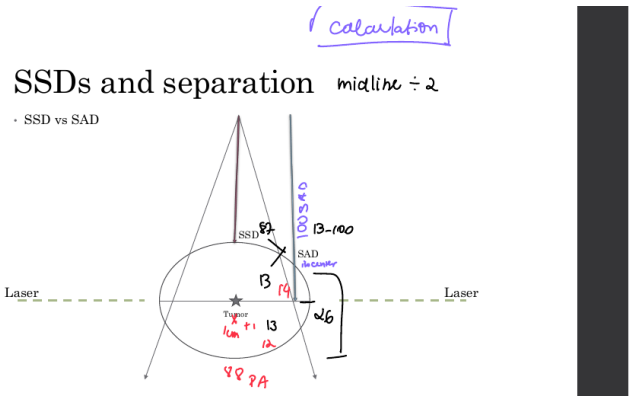
A bigger depth gives a smaller SSD. A smaller depth gives a bigger SSD. They always move opposite.

Separation

Separation is how thick the patient is along the beam line. An AP separation of 28 cm means 28 cm from the front skin to the back skin. A lateral separation goes from right skin to left skin.

Midplane means the ISO is exactly in the middle. So the depth from each side is **separation ÷ 2**. That is the "midline ÷ 2" note on slide 13.

Your slide 13



On the slide the patient is 26 cm. Half is 13. So SSD = 100 - 13 = 87. Then the tumor needed the ISO 1 cm lower (toward posterior). The top depth became 14 and the bottom depth became 12. So the PA SSD became 100 - 12 = **88**. That is the red "88 PA" in your notes.

2. The separation trick

This is the part that confuses you, so let us go really slow.

When the doctor says “shift the ISO 3 cm anterior”, the dot inside the patient moves 3 cm toward the **front skin**. Nothing else moves. The patient stays the same size.

The golden rule

TOWARD a side that side's depth gets **smaller** that side's SSD gets **bigger**

AWAY from a side that side's depth gets **bigger** that side's SSD gets **smaller**

Why? If the dot walks closer to the front skin, there is less body between the front skin and the dot. Less depth. And 100 minus a small number is a big number.

Which side matches which SSD

Field name	Beam goes in through	Use this depth
AP SSD	the ANTERIOR skin (front)	depth from the front
PA SSD	the POSTERIOR skin (back)	depth from the back
RT LAT SSD	the RIGHT skin	depth from the right
LT LAT SSD	the LEFT skin	depth from the left

Tip: the **first letter** of AP is A, so the beam enters Anterior. The first letter of PA is P, so it enters Posterior.

The 5 steps, every time

Step 1. Write the separation.

Step 2. Divide by 2. That is the starting depth for both sides.

Step 3. Look at the shift word. The side it names gets **minus** the shift. The other side gets **plus** the shift.

Step 4. Check. The two depths must add up to the separation again.

Step 5. SSD = 100 - depth for the side the question asks.

All 4 shifts in one place

Shift **anterior** front depth minus, back depth plus AP SSD goes up, PA SSD goes down

Shift **posterior** back depth minus, front depth plus PA SSD goes up, AP SSD goes down

Shift **left** left depth minus, right depth plus LT LAT SSD goes up, RT LAT SSD goes down

Shift **right** right depth minus, left depth plus RT LAT SSD goes up, LT LAT SSD goes down

Secret check trick

The two opposite SSDs always add up to **200 - separation**.

Example from Q2. 80.5 + 85.5 = 166. And 200 - 34 = 166. It matches, so your answer is right. If it does not match, you made a mistake somewhere.

Mistakes to avoid

Do not do **100 - shift**. The shift (like 2.5) is only how far the dot moved. It is not the depth.

Do not forget to divide by 2 first.

Read the word carefully. Toward ANTERIOR changes the front depth. Not the back one.

3. Separation Worksheet #2, solved

In every drawing the pink side is where the AP (or RT LAT) beam goes in. The mint side is where the PA (or LT LAT) beam goes in. The gold arrow shows how the ISO moved.

Question 1 and 1a

28 cm AP. ISO at midplane. Find AP SSD and PA SSD.



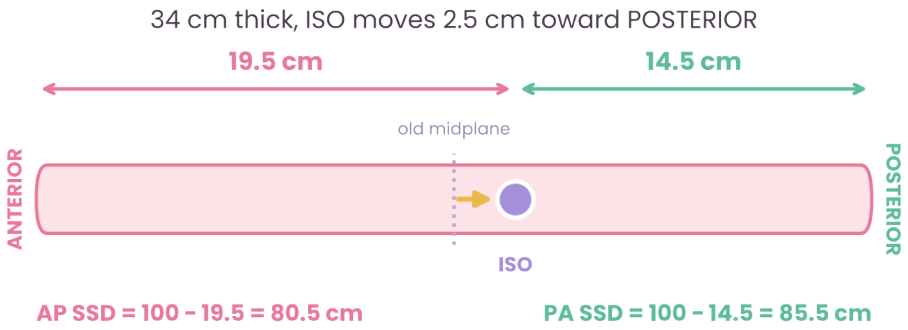
- 28 ÷ 2 = 14 cm from the front and 14 cm from the back.
- No shift, so nothing changes.
- AP SSD = 100 - 14 = **86 cm**
- PA SSD = 100 - 14 = **86 cm**
- Check 86 + 86 = 172 = 200 - 28

About your work

Your answers 86 and 86 are correct. At midplane both SSDs are always the same.

Question 2 and 2a

34 cm AP. Midplane. Shift ISO 2.5 cm **posterior**.



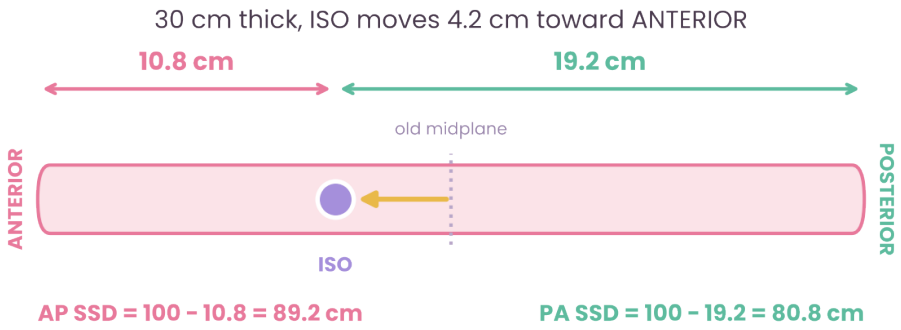
- 34 ÷ 2 = 17 cm on each side.
- Toward posterior, so the back depth gets minus. 17 - 2.5 = 14.5 cm.
- The front depth gets plus. 17 + 2.5 = 19.5 cm.
- Check 19.5 + 14.5 = 34
- AP SSD = 100 - 19.5 = **80.5 cm**
- PA SSD = 100 - 14.5 = **85.5 cm**

About your work

Your answers 80.5 and 85.5 are correct. You found the back depth first and did 34 - 14.5 to get the front. That works great.

Question 3 and 3a

30 cm AP. Midplane. Shift ISO 4.2 cm **anterior**.



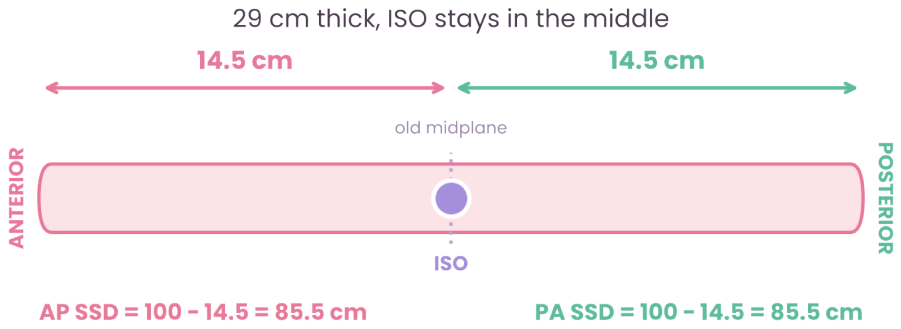
- 30 ÷ 2 = 15 cm on each side.
- Toward anterior, so the front depth gets minus. 15 - 4.2 = 10.8 cm.
- The back depth gets plus. 15 + 4.2 = 19.2 cm.
- AP SSD = 100 - 10.8 = **89.2 cm**
- PA SSD = 100 - 19.2 = **80.8 cm**
- Check 89.2 + 80.8 = 170 = 200 - 30

About your work

Your final answers are correct. But look at your line for 3. You wrote "100 - 19.2 = 89.2". That math is wrong on paper (100 - 19.2 is 80.8). The right line is 100 - 10.8 = 89.2. Fix that line so the teacher does not take points.

Question 4 and 4a

29 cm AP. Midplane.



- 29 ÷ 2 = 14.5 cm on each side.
- AP SSD = 100 - 14.5 = **85.5 cm**
- PA SSD = 100 - 14.5 = **85.5 cm**

About your work

Correct. Both 85.5.

Question 5 and 5a

45 cm from left to right. Midplane. Shift ISO 7 cm to the **left**.



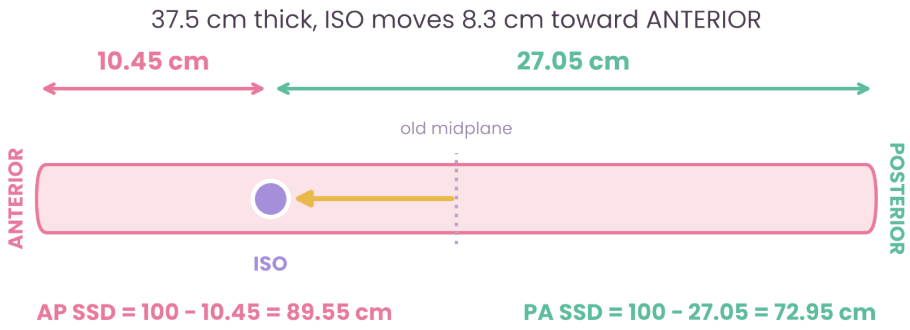
- Same rule, just right and left instead of front and back.
- 45 ÷ 2 = 22.5 cm on each side.
- Toward left, so the left depth gets minus. 22.5 - 7 = 15.5 cm.
- The right depth gets plus. 22.5 + 7 = 29.5 cm.
- RT LAT SSD = 100 - 29.5 = **70.5 cm**
- LT LAT SSD = 100 - 15.5 = **84.5 cm**
- Check 70.5 + 84.5 = 155 = 200 - 45

About your work

Both correct. Add “cm LT LAT SSD” next to 84.5 so it looks finished like the others.

Question 6 and 6a (blank on your sheet)

37.5 AP. Midplane. Shift ISO 8.3 cm **anterior**.



- The sheet says “37.5m” but it is a typo. It means 37.5 cm.
- 37.5 ÷ 2 = 18.75 cm on each side.
- Toward anterior, so the front depth gets minus. 18.75 - 8.3 = 10.45 cm.
- The back depth gets plus. 18.75 + 8.3 = 27.05 cm.
- AP SSD = 100 - 10.45 = **89.55 cm**
- PA SSD = 100 - 27.05 = **72.95 cm**
- Check 89.55 + 72.95 = 162.5 = 200 - 37.5

About your work

Write these on your sheet.

Question 7 and 7a (blank on your sheet)

22 cm AP. Midplane. Shift ISO 2.7 cm **posterior**.



- 22 ÷ 2 = 11 cm on each side.
- Toward posterior, so the back depth gets minus. 11 - 2.7 = 8.3 cm.
- The front depth gets plus. 11 + 2.7 = 13.7 cm.
- AP SSD = 100 - 13.7 = **86.3 cm**
- PA SSD = 100 - 8.3 = **91.7 cm**
- Check 86.3 + 91.7 = 178 = 200 - 22

About your work

Write these on your sheet. See how the PA SSD got bigger? The ISO moved toward the back.

Answer key

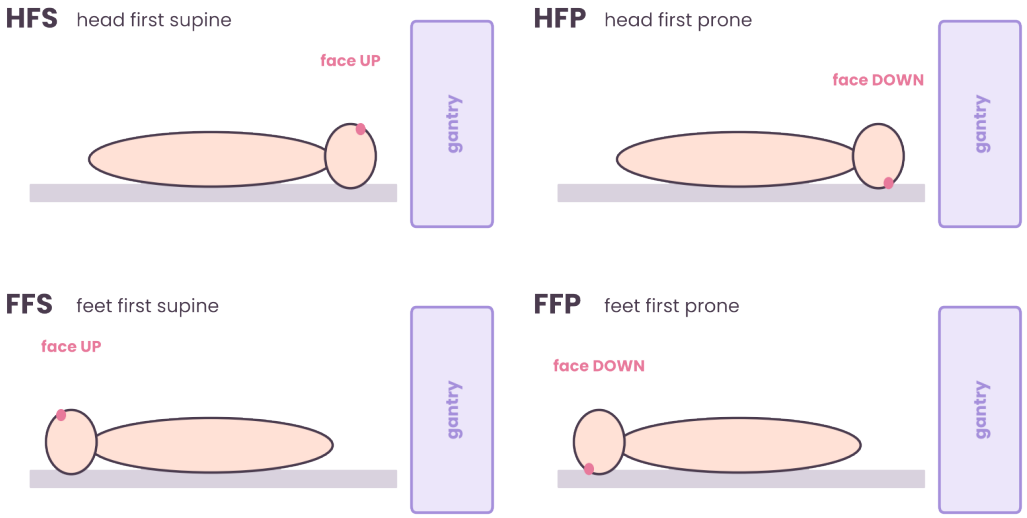
Q	AP or RT LAT SSD	PA or LT LAT SSD
1	86 cm	86 cm
2	80.5 cm	85.5 cm
3	89.2 cm	80.8 cm
4	85.5 cm	85.5 cm
5	RT LAT 70.5 cm	LT LAT 84.5 cm
6	89.55 cm	72.95 cm
7	86.3 cm	91.7 cm

4. Practice (cover the answers)

#	Question	Answer
1	40 cm AP, midplane. AP SSD and PA SSD?	20 and 20. Both SSDs are 80 cm.
2	32 cm AP, midplane, shift 3 cm posterior. AP SSD and PA SSD?	16 each. Front 19, back 13. AP 81 cm, PA 87 cm.
3	24 cm AP, midplane, shift 2 cm anterior. AP SSD and PA SSD?	12 each. Front 10, back 14. AP 90 cm, PA 86 cm.
4	36 cm lateral, midplane, shift 4 cm right. RT LAT SSD and LT LAT SSD?	18 each. Right 14, left 22. RT LAT 86 cm, LT LAT 78 cm.
5	27 cm AP, midplane, shift 1.5 cm anterior. AP SSD and PA SSD?	13.5 each. Front 12, back 15. AP 88 cm, PA 85 cm.
6	The ISO moves anterior. Which SSD gets bigger?	AP SSD. The front depth got smaller.
7	AP SSD is 84 and separation is 30. What is PA SSD?	200 - 30 = 170. 170 - 84 = 86 cm. (Secret check trick!)

5. Patient positions

The first two letters tell you what goes into the machine first. The last letter tells you if the face looks up or down.



Code	Means	Remember
HFS	head first supine	head into the gantry, face up. The normal one.
HFP	head first prone	head into the gantry, face down
FFS	feet first supine	feet into the gantry, face up
FFP	feet first prone	feet into the gantry, face down

Memory trick

Supine has the word “up” inside the sound (s**up**ine). Face up.

Prone sounds like “pro” going down a slide on the belly. Face down.

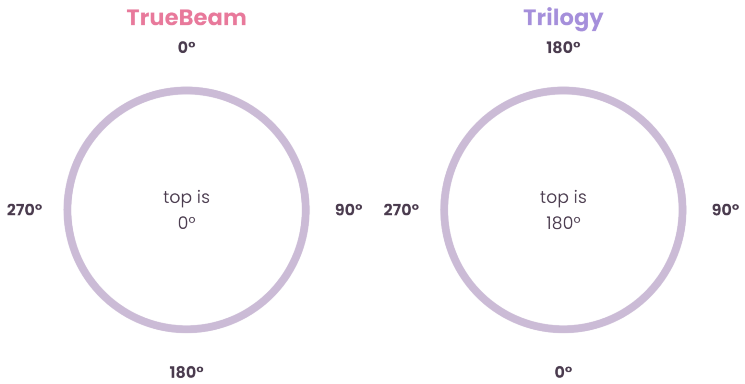
What the field names mean

Name	Full name	Beam goes in from
AP	anterior posterior	the front
PA	posterior anterior	the back
RT LAT	right lateral	the patient's right side
LT LAT	left lateral	the patient's left side
RAO	right anterior oblique	between front and right
LAO	left anterior oblique	between front and left
RPO	right posterior oblique	between back and right
LPO	left posterior oblique	between back and left

Always the **patient's** right and left. Never your right and left.

6. Gantry angles made simple

Picture yourself standing at the **foot of the couch**, looking at the machine. Your class sketches show the circle from there.



The only difference

Both machines have **90 on the right** and **270 on the left** of the circle.

Only top and bottom are different. **TrueBeam top = 0**, bottom = 180. **Trilogy top = 180**, bottom = 0.

Trick to remember. TrueBeam is the newer machine, it starts fresh at 0 on top. Trilogy is the older one and is upside down.

The flip trick (this is the big one)

First learn ONE picture, the HFS patient. With HFS, you see the face on top (anterior up) and the patient's **right side is on your left, at 270**. Just like when you look at a person lying in front of you.

Then remember these flips.

- **Prone** flips the patient over. The front goes down, the back goes up, and right and left swap.
- **Feet first** turns the patient around. Front stays up, but right and left swap.
- **Feet first prone** has two swaps, so right and left go back to the HFS sides. Only front and back flip.

Position	Patient top (12 o'clock)	Patient RIGHT is at	Patient LEFT is at
HFS	Anterior	270 side	90 side
FFS	Anterior	90 side	270 side
HFP	Posterior	90 side	270 side
FFP	Posterior	270 side	90 side

How to answer ANY oblique question in 4 steps

1. Draw the circle for the machine (TrueBeam 0 on top, or Trilogy 180 on top).

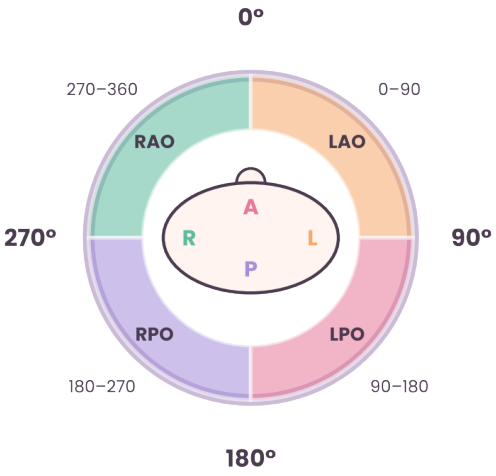
2. Draw the patient inside. Write A and P (top or bottom). Write R and L (using the flip table).

3. Find the field. RAO is between A and R. LPO is between P and L. And so on.

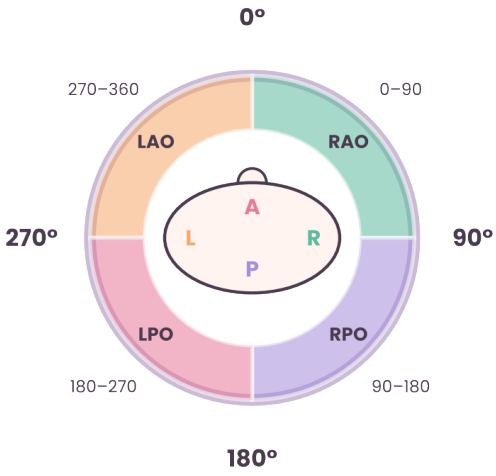
4. Read the numbers on the two ends of that quarter. Any angle between them is that oblique. A number near the middle (45 away from the ends) is the classic choice.

All the TrueBeam answers in one page

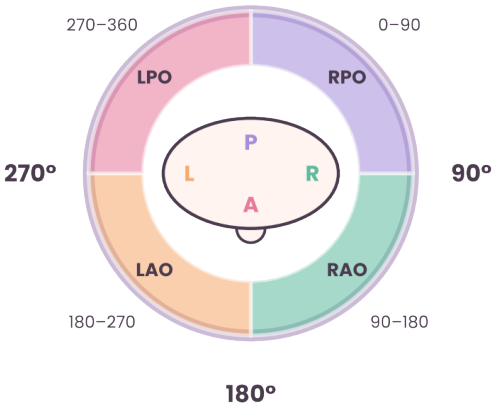
TrueBeam HFS



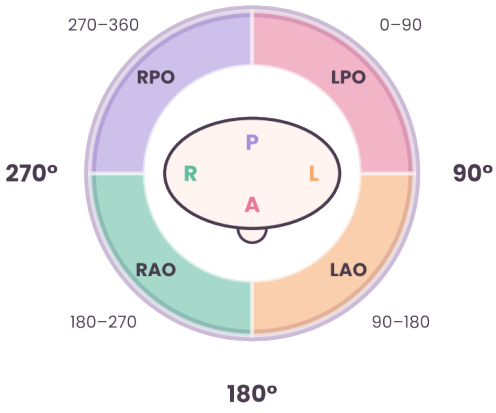
TrueBeam FFS



TrueBeam HFP

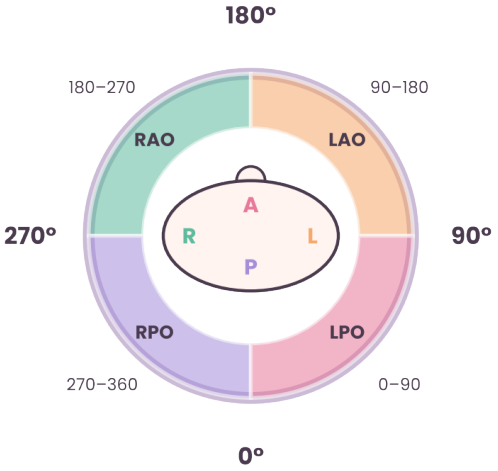


TrueBeam FFP

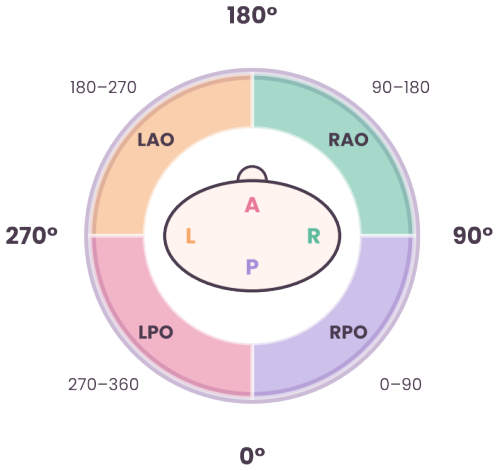


All the Trilogy answers in one page

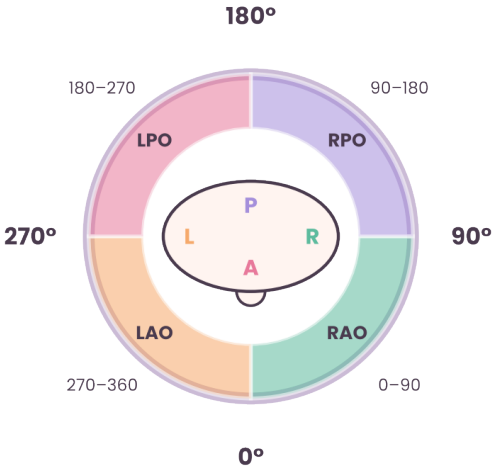
Trilogy HFS



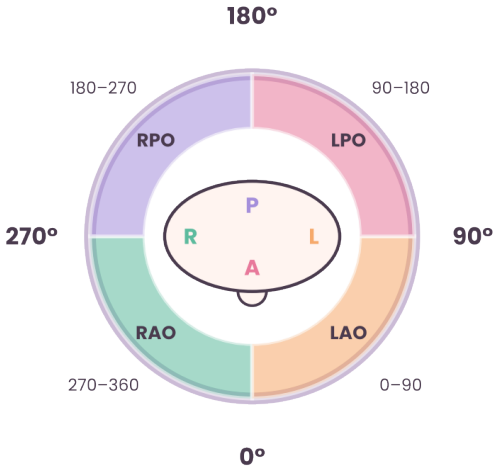
Trilogy FFS



Trilogy HFP

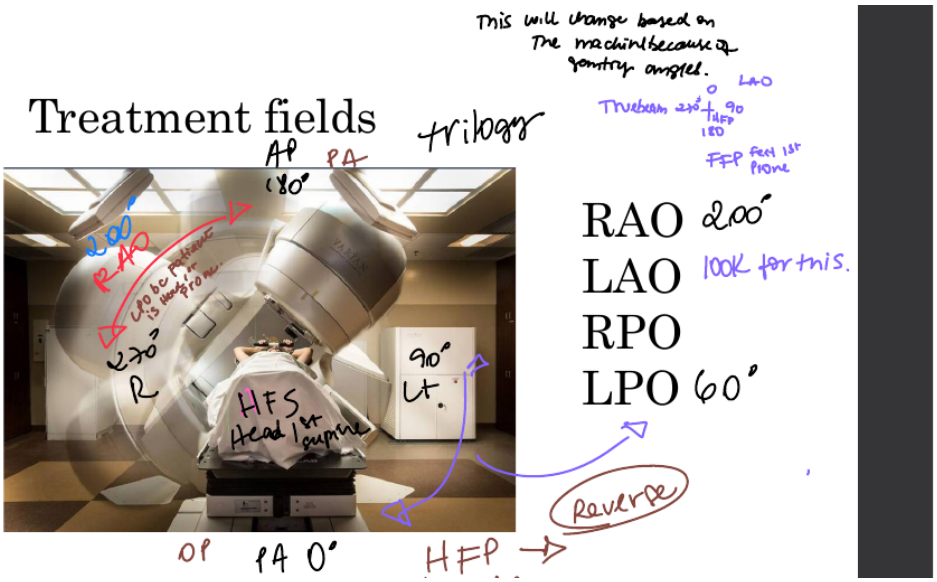


Trilogy FFP



Check with your slide 7. Trilogy HFS shows **RAO 200** and **LPO 60**. Both fit the HFS circle above. The slide also says the 180 to 270 arrow becomes **LPO** when the patient is head first prone. That matches the HFP circle.

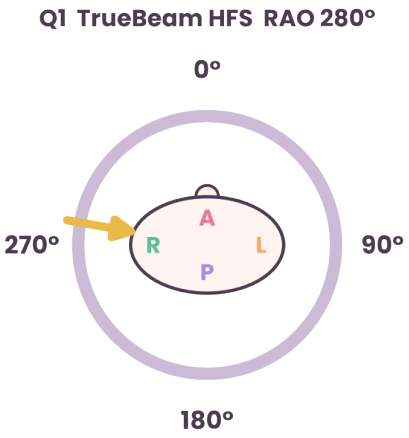
Your slide 7 notes



Your teacher wrote “Reverse” for HFP and “this will change based on the machine because of gantry angles”. That is exactly the flip trick plus the TrueBeam vs Trilogy difference.

7. Imaging and Positioning Worksheet #2

Gold arrow = where the beam goes in. Every circle is drawn from the foot of the couch.

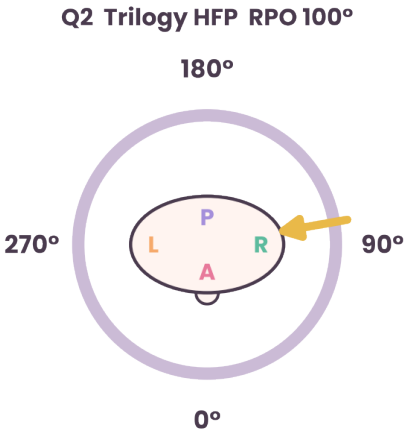


Q1. TrueBeam, HFS, RAO

TrueBeam, so 0 on top. HFS, so anterior is on top and right is at 270.

RAO sits between 0 (front) and 270 (right). So any angle from **270 to 360**.

Your answer **280°** fits . A classic RAO would be 315.

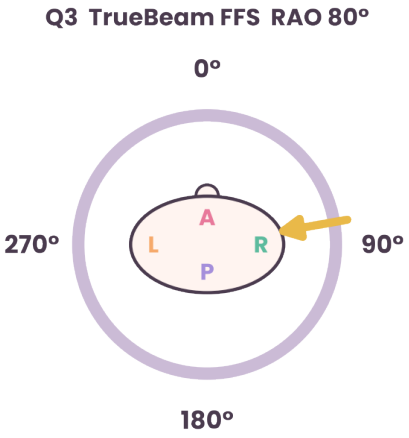


Q2. Trilogy, HFP, RPO

Trilogy, so 180 on top. Prone, so the back is on top and right moved to 90.

RPO sits between 180 (back) and 90 (right). So **90 to 180**.

Your answer **100°** fits .



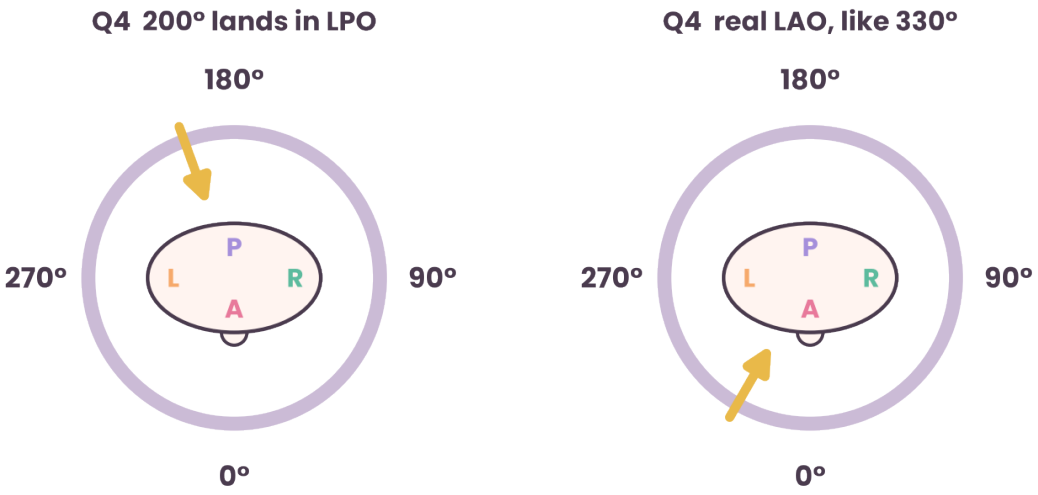
Q3. TrueBeam, FFS, RAO

TrueBeam, 0 on top. Supine, so front on top. Feet first, so right swapped to 90.

RAO sits between 0 (front) and 90 (right). So **0 to 90**.

Your answer **80°** fits .

Q4. Trilogy, HFP, LAO (please double check this one)



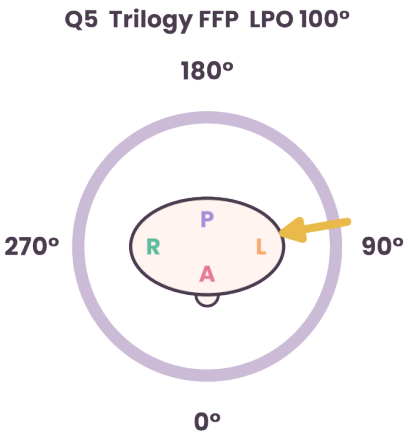
Trilogy, so 180 on top. Prone, so the back is on top and the front is at the bottom (0). Head first prone, so right is at 90 and left at 270.

LAO is between the front (0) and the left (270). So the LAO angles are **270 to 360**, like 300 or 330.

Your sheet says 0 to 270 and picks **200°**. But 200 is between 180 (back) and 270 (left). That is the **LPO** spot, not LAO. Your own slide 7 says the same thing. On the Trilogy picture the 180 to 270 arrow is written as "LPO because patient is head first prone".

My honest note

By the rule, LAO for Trilogy HFP should be about **330°** (anything from 270 to 360). Ask your teacher before 11/12 if 200 was the answer given in class. If it was, ask why, because it does not match slide 7.



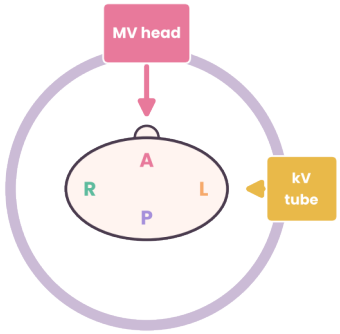
Q5. Trilogy, FFP, LPO

Trilogy, 180 on top. Prone, so the back is on top. Feet first prone means two swaps, so right is back at 270 and left at 90.

LPO sits between 180 (back) and 90 (left). So **90 to 180**.

Your answer **100°** fits . Notice Q2 and Q5 are both 100 but they are different fields. The feet first flip did that.

kV questions (6, 7, 9)

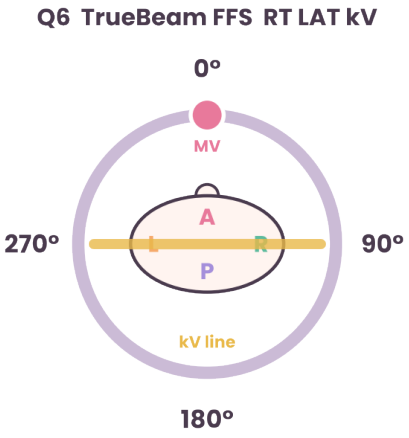


Same gantry angle, two beams.
kV comes from 90° away from MV.

The treatment beam (MV) and the imaging tube (kV) are on the same gantry, but the kV tube sits **90° away** from the MV head. Look at the slide 2 photo. The kV arms stick out to the sides.

So the trick is backwards from MV.

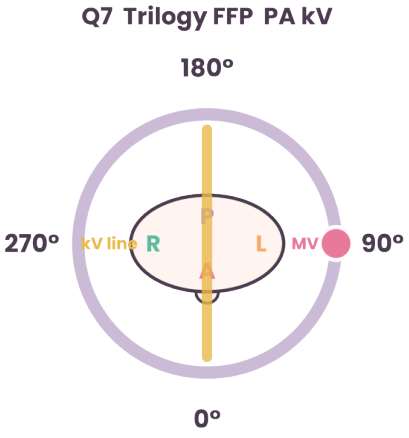
- Want a **kV lateral**? Park the gantry where the MV would shoot AP or PA (top or bottom).
- Want a **kV AP or PA**? Park the gantry at a lateral angle (90 or 270).



Q6. TrueBeam, FFS, RT LAT kV

For a MV right lateral, FFS right is at 90, so MV = 90 (you wrote that too).

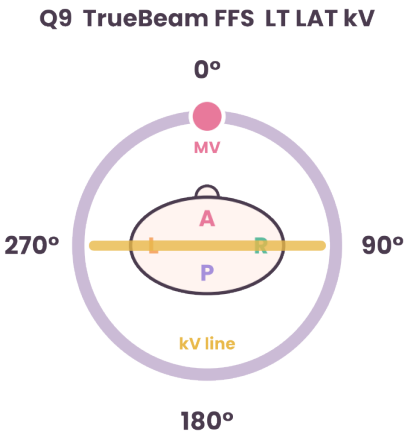
For the kV, move 90 away. The gantry goes to the top. **kV = 0°**



Q7. Trilogy, FFP, PA kV

PA is a top/bottom picture, so the kV needs the gantry at a side.

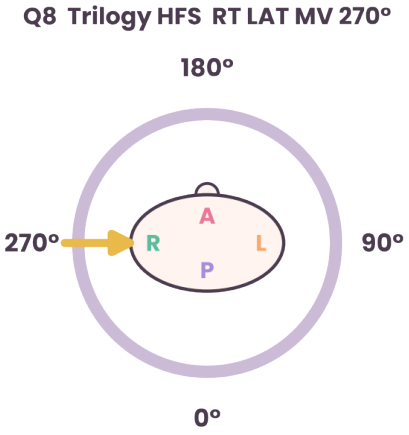
Your answer **90°** puts the MV head at the side and the kV line goes front to back.



Q9. TrueBeam, FFS, LT LAT kV

A lateral kV again, so the gantry goes on top. **0°**

Notice Q6 and Q9 are both 0. In class, a kV lateral at 0 counts for right or left. Which side the kV tube is on decides the exact one. If your teacher wants the strict side, the other answer would be 180. Good question to ask.



Q8. Trilogy, HFS, RT LAT MV

This is MV, so no 90 trick. The beam comes straight from the head.

HFS right is at 270. So **270°**

Worksheet #2 summary

Q	Machine and position	Field	Range	Your answer
1	TrueBeam HFS	RAO	270 to 360	280
2	Trilogy HFP	RPO	90 to 180	100
3	TrueBeam FFS	RAO	0 to 90	80
4	Trilogy HFP	LAO	270 to 360	200 (that is LPO). Try 330
5	Trilogy FFP	LPO	90 to 180	100
6	TrueBeam FFS	RT LAT kV	gantry top	0 (MV would be 90)
7	Trilogy FFP	PA kV	gantry side	90
8	Trilogy HFS	RT LAT MV	exact	270
9	TrueBeam FFS	LT LAT kV	gantry top	0 (ask about 180)

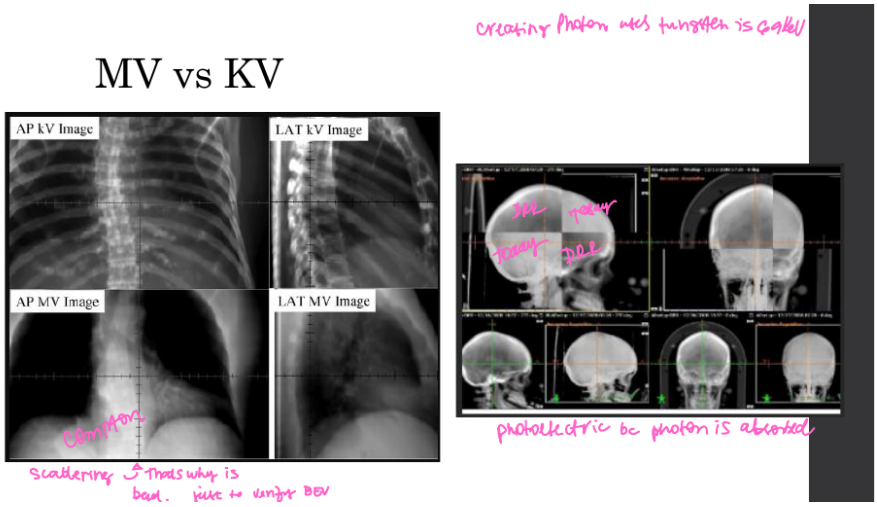
8. The rest of the class slides

On Board Imaging (slide 2)

OBI means On Board Imaging. Your note says the whole thing is OBI, and you use the buttons on the side panel. There are 3 kinds of pictures.

Type	2D or 3D	What it is	Best for
kV	2D	a normal x ray from the kV tube on the side arm	**bone**, very clear
MV	2D	a picture made with the treatment beam itself	checking the field shape (BEV), like the **flash** on a breast tangent
CBCT	3D	cone beam CT. The gantry spins and builds slices	**soft tissue**

MV vs kV (slide 3)



Why does kV look so sharp and MV so gray? It is the physics from your notes.

- **kV** is low energy. The main interaction is the **photoelectric effect**. The photon gets absorbed. Bone absorbs a lot more than soft tissue, so bone pops out white. Great contrast.
- **MV** is high energy. The main interaction is **Compton scattering**. Bone and soft tissue look almost the same, so the image looks blurry and gray. But it is the real treatment beam, so it is used to verify the BEV (beam's eye view) and the MLC shape.

Your notes also mention that the target that makes the photons is tungsten. In the right picture, the checkerboard mixes the **DRR** (from the planning CT) with **today's** image so you can see if they line up.

CBCT registration (slide 4)

Registration = image matching. You take today's CBCT and lay it on top of the planning CT. If the bones or organs do not line up, the couch is shifted to fix it.

Your notes on **MV CBCT**. It gives more exposure (6.5 cGy on the slide), it has more scattering, so lower image quality. But it is good for patients with **metal**, because MV does not make big metal artifacts.

Which is which? (slide 5)

The picture shows two beams coming out of the gantry, a red one and a yellow one, and arrows showing the gantry turning. One is the MV treatment beam from the head and one is the kV imaging beam from the side arm. The point of the slide is the same as the kV trick. They are **90° apart** and the whole thing spins together for a CBCT.

What structures are seen better (slide 6)

Your notes say kV = **bone**, CBCT = **soft tissue**, and MV is used for **flash on a breast tangent**. On the MV image you marked the heart, lung, chest wall, breast and the flash (the part of the field that goes past the skin into air).

Full vs empty (slides 9 and 10)

The bladder and rectum change size every day. A full bladder pushes the bowel up and out of the field. A full rectum can push the prostate or the uterus to a new spot. For prostate you marked the prostate and seminal vesicles. For the female pelvis you marked the uterus moving with a full bladder.

The goal is to be the **same every day** as the planning CT. Your note says 2 bottles of water 45 minutes before. That is the class example, every clinic has its own rule.

Breath hold vs free breathing (slide 11)

DIBH means deep inspiration breath hold. The patient takes a big breath and holds it. The lungs fill up, and the heart moves down and away from the chest wall. So for a **left breast**, the beam hits less heart. Your note says on the DRR you see the ribs, it is 2D.

Belly board (slide 12)

Used with the patient **prone**. The belly drops into the hole, and gravity pulls the small bowel forward and away from the pelvis field. So less bowel gets dose. Your note says “belly board push”. Your teacher also wrote practice sectional anatomy, so review the pelvis slices.

9. Cheat sheet

Separation

SSD = 100 - depth. Midplane depth = separation ÷ 2.

Shift TOWARD a side that depth minus the shift that SSD bigger.

The other side depth plus the shift SSD smaller.

Check. Two depths add to the separation. Two SSDs add to 200 - separation.

Angles

TrueBeam top 0. Trilogy top 180. Both 90 right and 270 left.

HFS. Front up, patient right at 270.

Prone flips front/back AND right/left. Feet first flips right/left only. FFP = right/left back to normal.

Oblique = the quarter between its two letters. RAO = between A and R.

kV is 90° from MV. kV lateral gantry top or bottom. kV AP/PA gantry at a side.

Imaging

kV = bone (photoelectric). MV = verify field/flash (Compton, gray). CBCT = 3D soft tissue.

Registration = image matching. MV CBCT good for metal, more dose.

Full bladder pushes bowel away. Be the same every day.

DIBH moves heart away from left breast. Belly board = prone, bowel falls away.

You got this, Fran. Draw the bar, draw the circle, then answer.